Performance of Work Statement (PWS)
For
Condition Based Maintenance Plus (CBM+)

1.0 Scope.

This Performance of Work Statement (PWS) sets forth the work efforts required to procure and integrate data loggers with Marine Corps' ground vehicle platforms to collect Controller Area Network (CAN) data, transfer data over approved networks, store, catalog, and manage data in government cloud repositories, apply Artificial Intelligence (AI) and Machine Learning (ML) models and other data analytic techniques to analyze the data, and visualize the data in an easy to understand decision support tool so Marines can act on the information All associated logistics, technical information and documentation for hardware and software delivered to be delivered are documented in this PWS. The CBM+ capability consists of the necessary hardware, software, data analytics (AI/ML), data visualizations (dashboards), interfaces, and all associated buses, connectors, and cabling. The contractor shall provide the requisite program management, engineering, test and evaluation and logistics support to ensure that delivery schedules, performance requirements and overall supportability of the CBM+ system are accomplished as set forth in the contract. The Contractor is responsible for providing all materials, services, and necessary support documentation needed to complete the tasks identified in this PWS.

2.0 Applicable Documents.

The following documents form a part of this PWS to the extent specified herein. The most recent version of the referenced document at the time of the contract shall be used unless otherwise specified. In the event of a conflict between the applicable documents and this PWS, the PWS shall take precedence. All second tier and below references cited in mandatory compliance shall be considered as guidance only. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

2.1 Military Standards and Specification – Mandatory Compliance.

MIL-PRF-28800F	Test Equipment for Use with Electrical and Electronic Equipment General Specification For
MIL-STD-129	Military Marking for Shipment and Storage
MIL_STD-130	Identification Marking of U.S. Military Property
MIL-STD-882	Standard Practice for System Safety
MIL-STD-2073-1	DOD Standard Practice for Military Packaging
MIL-STD-348B	Department of Defense Interface Standard: Radio Frequency Connector Interfaces
MIL-PRF-38793C	Technical Manuals; Calibration Procedures – Preparation
MIL-PRF-29612B	Training Data Products

DoD 8510.01	Risk Management Framework (RMF) for DoD Information Technology
MCSCO 9300.1	Marine Corps Systems Command Implementation of the Naval Lithium Battery Safety Program
NAVMC 1553.1A	Marine Corps Systems Approach to Training, User Guide

2.2 Military Standards and Specifications – Guidance Only.

MIL-STD-1686	Electrostatic Discharge Control Program for Protection of Electrical and Electronics Parts, Assemblies, and Equipment

2.3 Federal Standards – Mandatory.

DFARS252.245-7005	Management and Reporting of Government Property
FED-STD-5	Standard Guides for Preparation of Proposed Item Logistics DataRecords
FED-STD-313	Material Safety Data, Transportation Data, and Disposal Data for Hazardous Materials Furnished to Government Activities
GEIA-HB-0007	Logistics Product Data Handbook

2.4 Handbooks – Guidance Only.

MIL-HDBK-773A	Electrostatic Discharge Protective Packaging
MIL-HDBK-1839A	Handbook Calibration and Measurement Requirements
MIL-HDBK-29612-4A	Glossary for Training
MIL-HDBK-502	Product Support Analysis

2.5 Other Government Documents.

Unless otherwise stated, the following documents may be obtained from the Defense Logistics Agency Document Services, Building 4/D, 700 Robbins Avenue, Philadelphia, PA 19111-5094, or visit http://dodssp.daps.mil.

DOD 4100.39M, Vol. 6	Federal Logistics Information System
DOD 4120.3-M	Defense Standardization and Specification Program Policies, Procedures, and Instructions

2.6 Non-Government Documents.

A-A-50271	Plate, Identification
ASME Y14.34	Associated Lists
ASME Y14.100	Engineering Drawing Practices

GEIA-STD-0007 Logistics Product Data

(Copies of American Society of Mechanical Engineers (ASME) documents are available from www.asme.org or American Society of Mechanical Engineers Information Central Orders/Inquires, P.O. Box 2300, Fairfield, NJ 07007-2300.)

ASTM D 3951 Standard Practice for Commercial Packaging

ASTM D4169-09 Standard Practice for Performance Testing of Shipping Containers and Systems

(Copies of ASTM documents are available from www.astm.org or American Society of Testing and Materials International, 100 Barr Dive, West Conshohocken, PA 19428-2959.)

EIA-625 Requirements for Handling Electrostatic Discharge-Sensitive (ESDS) Device

EIA-649 National Consensus Standard for Configuration Management

(Copies of IEEE/EIA are available from www.ieee.org or Institute of Electrical and Electronics Engineers Service Center, 445 Hoes Lane, Piscataway, NJ 08854-1331.)

NAS 411 Hazardous Materials Management Program

3.0 Requirements.

The Contractor shall perform all tasks required and delineated in this PWS to deliver the CBM+ capability and associated logistics information and technical documentation.

3.1 Administrative and Program Management.

3.1.1 Program Management. The Contractor shall establish and maintain program management practices throughout the period of performance. Program management practices shall provide visibility into the Contractor's organization and techniques used in managing the program performance, cost, schedule, subcontractor(s), and data management and deliverables. Documentation shall be readily available to Government representative(s) whenever requested.

3.1.2 Subcontractor Management. The Contractor is responsible for performance of requirements delineated in this PWS and shall institute appropriate management actions relative to subcontractor performance. Requirements that are contractually specified shall apply to subcontractor performance; however, the Contractor shall be accountable for compliance by the sub-contractors and is responsible for ensuring all deliverable products comply with the contract requirements.

3.1.3 Data Management. The Contractor shall establish a single centralized system for management of all data required under this contract. Specific data management functions shall include: schedule control for deliverables, maintenance of deliverables, approval of deliverable format, and distribution and delivery of data products. An acceptable system for managing data would be web-based URL for access to documentation by the Government. The Government reserves the right to review and access all data associated with and developed for the CBM+ program.

3.1.4 Integrated Master Plan and Integrated Master Schedule. The Contractor shall develop and submit a detailed Integrated Master Plan (IMP) and Integrated Master Schedule (IMS). The IMP shall include the major events, accomplishments, and criteria required to complete the project, including: milestones, program reviews, Contract Data Requirements List (CDRL) deliverables, builds, test events, training, production, fielding to/installing at designated locations, sustainment activities, legacy system retirement, delivery orders, and other elements as needed to address the entire scope of work. The IMS shall be an integrated, networked schedule containing all the discrete work packages and planning packages, or lower-level tasks, necessary to support the events, accomplishments, and criteria as detailed in the IMP.

(CDRL B001, Integrated Program Management Data and Analysis Report (IPMDAR))

3.1.4.1 After the Government's review and approval of the first IMS submission, the IMS baseline will be agreed upon by the Government and Contractor. Changes to the baseline schedule shall be made in accordance with (IAW) the Contractor's schedule change management process. Any movement of contractual milestones in the baseline schedule shall be derived only from either authorized contract changes or an approved over target schedule. The Contractor shall provide updates to the IMS and coordinate with the Government to identify and establish linkages (i.e., task dependencies and/or work relationships) between Government and Contractor schedules. The Contractor shall coordinate with the Government to identify and resolve schedule and performance concerns, risks, and issues. The Contractor shall perform appropriate analyses of the IMS tasks, report potential or existing problem areas to Government team, and recommend corrective actions to eliminate or reduce schedule impact. The Contractor shall identify and report the Critical Path and Near Critical Path. The Government may require separate submission(s) of subcontractor IPMDARs.

(CDRL B001, Integrated Program Management Data and Analysis Report (IPMDAR))

3.2 Meetings, Formal Reviews, Conferences, and Audits.

3.2.1 The Contractor shall plan, host, coordinate, support, and conduct meetings, formal reviews, and conferences. The reviews shall be conducted at Government and Contractor facilities. All such reviews shall be included in the program schedule. The Contractor shall prepare and distribute all conference presentation materials and agendas for Government review and approval prior to the conference. The Government reserves the right to cancel any review or to require any review to be scheduled at critical points during the period of performance. A summary of all action items, responsible parties, and estimated completion dates shall be provided in the conference minutes. The Contractor shall prepare a monthly status report that includes all work performed during the reporting period and a status of remaining funding with funding expended to date. A consolidated action items listing shall be submitted as part of the Contractor's progress, status, and management report and shall also include status updates comprised of status update summaries, open and closed. Action item documentation, assignment of responsibility for completion, and due dates shall be determined prior to adjournment of all reviews. The contractor shall plan/host quarterly in progress reviews with the government Team to communicate all work performed to date.

(CDRL A001, Conference Agenda)
(CDRL A002, Conference Minutes)
(CDRL A003, Monthly Status
Report)

3.2.2 Post Award Conference. The Contractor shall host a Post Award Conference (PAC) at the Contractor's facility within 30 days after Contract Award. The purpose of the PAC is for the Contractor to review and demonstrate to the Government the management procedures, provide progress assessments, review technical and other specialty area status, and establish schedule dates for near-term critical meetings/actions. The Contractor shall present management, key personnel, and program implementation processes. The Contractor shall develop a conference agenda(s). The Contractor shall record/take conference minutes.

(CDRL A001, Conference Agenda)
(CDRL A002, Conference Minutes)

3.2.3 Weekly Integrated Product Team (IPT) meeting. The contractor shall attend weekly virtual integrated product team meetings and be available on a as needed basis to meet with the Government to discuss program requirements, status, schedule, budget, issues and other topics relevant to the CBM+ program.

3.2.4 Provide responses to Government inquiries within one business day.

3.2.5 Obtain a Common Access Card (CAC) for employees that require access to Military Bases and other DoD facilities. Contractor personnel who require regular access to MARCORSYSCOM Buildings will obtain an access badge and always display it when in the facilities. Additionally, contractor personnel who require authorized access to Marine Corps networks will submit a SAAR and comply with all training and security requirements for access. Personnel requiring a privileged level of access to Marine Corps networks to perform work associated with this PWS will comply with all additional stipulations required. Including but not limited to appointment as a Cyber Security Workforce Professional.

3.3 Data Analysis, Visualizations, and Pipelines

3.3.1 The contractor shall provide qualified personnel who can gain privileged access to Government cloud computing environments on the Marine Corps Enterprise Network (MCEN) in the form of Microsoft Azure and on the Non-Classified Internet Protocol Router Network (NIPR) in the form of the Navy's Jupiter enclave in the Advana cloud. Privileged (administrator) level access on the MCEN requires Security Plus certification. Waivers to this requirement can be obtained for qualifying degrees. Each environment contains the required data analytic and visualization tools required in performance of this PWS. The contractor shall use, build upon, and enhance currently available AI/ML models and dashboards in the environments.

3.3.2 The contractor shall conduct an analysis of current dashboards in the Government cloud environments and the data Extract, Transform, and Load (ETL) pipeline to produce a roadmap of future features and improvements required

3.3.3 The contractor shall provide data scientists, data engineers, and data visualization/user interface design expertise to analyze and visualize data collected from Marine Corps platforms that are fully qualified and meet MCEN standards for the privileged access as required to configure the Government provided cloud and networking environments.

3.3.4 The contractor shall support current systems, dashboards, policies, and procedures of the current data pipeline from the data entry point until the point of entry to Jupiter Advana. Tasks include but are not limited to:

- System administration tasks (Operating System Maintenance, user management, password resets etc.)
- Software updates in line with the configuration management plan and cyber security posture
- Wireless Access Point (WAP) heat mapping and troubleshooting
- Transition the current pipeline and support the transition of the WAPs to the MCEN

- Maintain and deliver source code; new, reused, or modified code to facilitate the data pipeline and dashboards IAW CDRL A005, Software Documentation and Source Code
- Test planned dashboard, model, and source code releases to ensure compatibility with current systems
- Ensure dashboards in the MS Azure and Jupiter Clouds reflect accurate information to Marine Commanders and Maintainers
- Perform cyber security scans and Security Technical Implementation Guides (STIG) reviews in support of annual security reviews and Authority to Operate (ATO) attainment/sustainment
- Maintain a CBM+ lab environment with field representative hardware designed to test updates and/or new cyber requirements prior to deploying new features
- The contractor shall deliver two major enterprise dashboard releases in a 12-month period. Major dashboard releases shall include enhance feature sets and new AI/ML models that improve the ability to implement predictive maintenance actions. Continuous releases that are part of the DevSecOps software development process that improve only backend functions are not counted as a major release. Dashboards for CBM+ and FARS shall communicate easy to understand maintenance information and bulk fuel status to multiple stakeholders. The enterprise level dashboards shall include the necessary feedback loop for user input. CDRL A008
- The contractor shall deliver the Maintainer Dashboard hosted in the USMC Enterprise Azure Cloud that is easily accessible using a CAC and PKI authentication. The Maintainer Dashboard shall provide vehicle specific information for CBM+ enrolled platform (including bumper number, TAMCN, platform type, location, last check-in, time since last check-in, operational status, data logger memory used vs. memory capacity, and a listing of vehicle specific active faults with associated work package in the Technical Manual [TM] when available). The Maintainer Dashboard shall enable display of vehicle information from all CBM+ participating USMC motor pools. The dashboard shall enable users to sort data displayed by location, vehicle type, and bumper number. Users shall also have the ability to search by TAMCN.

3.3.5 System Documentation. The contractor shall use industry standard configuration management practices and prepare and deliver software documentation for version control per Configuration Management Requirements for Defense Contractors EIA-649-1

3.3.6 The contractor shall deliver or update system documentation including but not limited to the below list IAW CDRL A006- Software Design Description

- System Baseline
- System Architectures
- System Specifications
- Test plans and procedures
- Wiki pages, Jira entries, and Jupyter notebooks
- Training documentation
- Software bill of Materials (SBOM) IAW CDRL A006 Software Design Description.

3.3.7 Current Environment Support. The contractor shall support the current Microsoft Azure and Jupiter Advana cloud environments. Tasks include but are not limited to:

- Jira Task Tracking
- Wiki management and upkeep
- On-boarding of contractor personnel to the environments
- Accuracy of the maintainer (Azure) and enterprise (Jupiter) level dashboards
- Provide input to data sharing agreements and data governance policies
- Daily monitoring of processes during normal business hours to proactively identify problems
- Optimization of the cloud architecture to ensure scalability

3.3.8 Additional Data Sources. The contractor shall incorporate additional data sources into the CBM+

pipeline and data visualizations. Data integration shall include analysis of potential predictive capabilities and benefits to the Marine Corps. Data sources and types may include but are not limited to the following:

- Fuel Automated Reporting System (FARS) data for fuel quantities, locations, and movement. FARS data shall be visualized on a separate and distinct dashboard in the Jupiter environment
- Equipment and parts history data and task notes from GCSS-MC
- Data collected by an on-platform computer system or at platform devices, like but not limited to the Electronic Maintenance Support System (EMSS)
- Fluid Analysis data from the Expeditionary Fluid Analysis System (EFAS).

3.3.9 New AI/ML models. The contractor shall incorporate additional predictive and analytical models for existing platforms in the Jupiter environment as data sources and platforms are added based on USMC priorities.

- Additional models shall include but is not limited to a Class IX parts predictive tool
- Additional platforms may include but are not limited to: The Amphibious Combat Vehicle (ACV), The Ultralight Tactical Vehicle (ULTV), The Armored Reconnaissance Vehicle (ARV), the High Mobility Artillery Rocket System (HIMARS), AMMPs Generators, and the Light Armored Vehicle (LAV).

3.3.10 Global Combat Support System-Marine Corps (GCSS). The contractor shall work with the GCSS-MC program office (Government and contractor personnel) to establish and support a bi-directional connection from the CBM+ data cluster in Jupiter to GCSS-MC. This connection will be used to update data fields in GCSS-MC to support predictive maintenance actions by Marines

3.3.11 Visual Integrated Tactical Logistics-Battle Management Aid (VITL-BMA). The contractor shall work with the Office of Naval Research, Logistics IT personnel and their contractors to establish a bi-directional connection with the VITL-BMA System.

3.4 Data Logger Hardware procurement, support, installation, and accountability. The contractor shall deliver up to 3,000 commercially available Technology Readiness Level 8 or higher CAN bus data logging hardware and ancillary items required for installation on Government designated Marine Corps platforms per ordering period. Hardware must meet all Performance Specification Requirements. Final quantities will be determined by funding levels.

3.4.1 Field Service Representative Support. The contractor shall provide on-site support for data logger and communications hardware installed by the CBM+ program at Camp Pendelton, CA, Camp Lejeune NC, and Okinawa, Japan. On-site support shall apply to current and any new hardware configurations installed. Tasks include but are not limited to:

- Bi-weekly touchpoints (physical or virtual) with USMC units
- Training on hardware and software
- Hardware monitoring
- Tier 1 support and escalation
- Install and de-install planning and coordination
- Interfacing between Marine Corps maintenance personnel, data scientists and software developers

3.4.2 Data logger installs and de-installs. The contractor shall conduct installs and de-installs of data logging hardware and communication equipment as needed on selected Marine Corps platforms on site at selected units. Locations include but are not limited to: Camp Pendelton, California, Camp Lejeune, North Carolina, and Hawaii, and Okinawa, Japan. If the scope of the install or de-install is beyond the level of onsite support the contractor shall provide additional personnel needed to accomplish the tasks. Install and de-install tasks include but are not limited to

- The contractor shall conduct site surveys to determine the level of effort required for installation and provide a detailed report to the program office and the unit prior to beginning installs IAW CDRL

A009, Site Survey Report

- The contractor shall provide a qualified team of installers commensurate to the level of effort required to complete the planned installations
- The contractor shall configure and kit all data logging hardware based on the unit provided platform availability list
- The contractor shall coordinate through the program office with selected units for platform availability and mitigations of scheduling conflicts
- The contractor shall account for data logging hardware and ship quantities to identified locations prior to installs
- The contractor shall have and execute a quality control plan IAW CDRL A010, Quality Control Plan which ensures all installs are performed to standards outlined in the installation guides and that they match the coordinated platform list.

3.4.3 Data logger storage and accountability. The contractor shall account for and store all data logger hardware until installed on USMC platforms.

3.5 Cyber Security Support. The contractor shall provide personnel to support cyber security efforts. These personnel shall draft cyber security related documents, conduct scans, assist with Security Technical Implementation Guides (STIG) implementation and perform other tasks required to navigate the risk management framework related to CBM+ which includes FARS data and data pipelines. This support includes but is not limited to

- Documenting security controls
- Creating and managing POAMs
- Updating authorization packages with hardware/software changes
- Conducing Annual Security Reviews (ASRs) and Reauthorizations.

3.6 Stakeholder Management. The contractor shall work in cooperations with other contracted performers in the program office, Naval Warfare Center (NSWC) Crane, Indiana Government and contracted personnel, Subject Matter Experts from Penn State University Applied Research Lab, Platform Program Offices, The Marine Corps Cyber Operations Group, Program Executive Office (PEO) Digital, PEO Land Systems and other personnel identified by the program office to accomplish the Marine Corps; objective of adopting CBM+ as a maintenance strategy.

3.7 Radio Frequency Interfaces and Connectors.

The Contractor shall ensure MIL-STD-348B radio frequency interfaces and connectors are utilized where applicable.

3.8 Environmental, Safety, and Occupational Health.

3.8.1 Systems Safety. The Contractor shall identify and evaluate safety and health hazards, identify risk levels, and establish a program that manages the probability and severity of all hazards associated with the manufacture, use, and disposal of the system IAW MIL-STD-882. Residual risks will be evaluated by the Government IAW guidance outlined in MIL-STD-882 and accepted as appropriate.

3.8.2 Safety Assessment. The Contractor shall document and prepare a Safety Assessment Report in accordance with MIL-STD-882E to identify procedural, hardware, and software related hazards that may be present in CBM+ data logger hardware, safety features of the hardware, software, and system design, and specific procedural controls and precautions that should be followed. The Contractor shall make recommendations applicable to hazards at the interface of the system to other systems. The minimum data shall include all identified ESOH hazards, the probability and severity of the hazard, the proposed of accomplished design/procedure to mitigate the risk, and the expected residual risks once the mitigation factors are implemented.

(CDRL A029, DI-SAFT-80102C, Safety Assessment Report)

3.8.3 Hazardous Material Management Program. The Contractor shall implement a Hazardous Materials Management Program (HMMP) IAW NAS 411. The Contractor shall avoid the use of toxic chemicals, hazardous materials, and ozone depleting substances in the design, operational support, and disposal of the CBM+ hardware. Manufacturing processes that will have a detrimental impact upon the environment shall be avoided. More information on chemicals and hazardous materials to avoid can be obtained from the Environmental Protection Agency (EPA). The Contractor shall deliver Material Safety Data Sheets to the Government IAW FED-STD-313.

(CDRL A004, Hazardous Material Management Program Plan)

3.9 <u>Quality Management System</u>.

The Contractor shall establish and maintain a Quality Management System (QMS) appropriate for the scope and complexity of the work performed under this contract. The QMS shall include documented procedures for configuration management, nonconforming control, corrective actions and quality planning. The system shall be based on widely recognized quality management principles such as those outlined in ISO 9001 or equivalent industry standards. The Contractor shall develop, maintain and deliver a Quality Assurance Surveillance Plan (QASP) describing the processes, inspections, controls, and corrective action procedures used to ensure all CBM+ deliverables meet contract Requirements.

(CDRL A006, Quality Assurance Surveillance Plan (QASP)

3.10 <u>Configuration Management</u>.

The Contractor shall maintain a configuration management (CM) process for the control of all hardware, software, firmware, configuration documentation, media, and parts representing or comprising the CBM+ system. The principles contained in MIL-HDBK-61 and EIA-649 may be used for guidance. The Contractor's CM process shall consist of configuration identification, configuration control, configuration status accounting, and configuration audits. The Contractor shall be responsible for any subcontractor's CM efforts.

3.10.1 Configuration Documentation. The Contractor shall develop a Software Version Description (SVD) to include installed software version numbers and software version release dates.

(CDRL A007, Software Version Description)

3.10.2 Software Application Program Interface (API) Documentation. The Contractor shall develop a Software Interface Control Document that includes the contractors API for interfacing with all CBM+ Hardware and Software. The document shall include all implemented functions and commands.

(CDRL A008, Software Application Program Interface (API) Documentation)

3.11 <u>Product Drawing Package</u>

3.11.1 The Contractor shall provide a product drawing package (PDP) that includes drawings/models for each component down to the piece part. The PDP shall include a control drawing for all Original Equipment Manufacturer (OEM) assemblies and components. If applicable, a parts list shall be included either on the relevant control drawing or as a separate drawing/document. All developed and provided drawings shall conform to ASME Y14.100 and ASME Y14.34.

(CDRL A009, Product Drawings/Models and Associated Lists)

3.12 Information Assurance

3.12.1 The Contractor shall ensure the CBM+ system is configured consistent with applicable DoD cybersecurity policies, standards, architectures, security controls, and validation procedures.

3.12.2 The Contractor shall perform a vulnerability scan of the CBM+ system and provide a compliance report.

(CDRL A024, Vulnerability Scan Compliance (VSC) Report)

4 PRODUCT SUPPORT & LOGISTICS REQUIREMENTS

4.0 Warranty

4.0.1 The design of the equipment shall be such that under normal use and operation, the equipment does not fail within 365 days of operation. The vendor must provide a failure analysis report demonstrating that the principal end item meets a Mean Time Between Failure (MTBF) requirement of at least 365 days.

(CDRL A005, Failure Summary and Analysis Report)

4.0.2 Standard Commercial Warranty. The Contractor shall provide a standard commercial warranty, for a minimum of 12 months starting 120 days after Government acceptance date, covering workmanship, materials, design, and all essential performance characteristics and requirements outlined in the CBM+ Hardware Performance Specification. The Contractor shall ensure that subcontractor and vendor warranties provide the same coverage and are passed through to the end item. The Contractor will pay for shipment from the unit to the repair facility and back to the unit for both CONUS and OCONUS locations.

4.1 Technical Manual (TM) Development

4.1.1 Commercial Manuals. The Contractor shall provide an OEM technical manual(s) for the data loggers. The Contractor shall identify copyrighted material IAW MIL-PRF-32216A and furnish the appropriate copyright release giving the Government permission to reproduce, update, or change the data contained in the OEM Technical Manual(s) for Government use only. When the Contractor uses a manual that covers another Contractor’s component(s) or a portion thereof and the Contractor’s manual contains copyrighted material, the Contractor shall be responsible for obtaining a copyright release from the Contractor and providing the copyright release to the Government.

(CDRL A010, OEM Technical Manual(s))

4.2 Training Products Development and Training Support

4.2.1 Commercial Training Products. The Contractor shall provide the Government with training materials, in the OEMs format, for operation and maintenance. Training products shall provide users with functionality, set-up, operating instructions, and maintenance instructions. Training products shall include any training materials in OEM Format, such as written/performance evaluations (with answer keys), job aids, videos, or other media. Training shall be consistent with procedures established in the appropriate TM. The combined operator and maintenance course shall be no more than forty (40) hours in duration with a 3:1 student to equipment ratio and 12:1 student to instructor ratio.

(CDRL A018, Commercial Training Products)

4.2.2 Systems Approach to Training (SAT) Training Products. The Contractor shall develop and format training materials for the training event IAW the NAVMC 1553.1A SAT User Guide. All training materials delivered to the Government shall be delivered in electronic Microsoft Office (Word, PowerPoint) format. Training will cover operation of CBM+ hardware and software.

4.2.3 Methods of Instruction. The preferred methods of instruction are presentation, demonstration, practical exercise, and hands-on training. The majority of training shall be practical exercise and hands-on training. Fault isolation and troubleshooting methods shall be accomplished by having students identify faulty components and/or software with particular emphasis on high-failure items.

4.2.4 Course Material. All course material shall be prepared per MIL-PRF-29612B and NAVMC 1553.1A. The Contractor shall provide a copy of all course material required to teach the course to each student attending Instructor and Key Personnel (I&KP) courses. The Contractor shall provide all training materials for each student taking the course. For further guidance, MIL-HDBK-29612-4A (1-5) may be used.

4.2.5 Lesson Plans. The Contractor shall develop CBM+ operator and maintainer lesson plans. The lesson plans shall include the Terminal Learning Objective (TLO) and Enabling Learning Objectives (ELOs). The lesson plans shall be detailed and comprehensive and not diverge/deviate from the technical manual(s).

(CDRL A019, Lesson Plans)

4.2.6 Student Outlines. The Contractor shall develop CBM+ operator and maintainer student outlines. The student outlines shall include the TLO and ELOs. The student outlines shall be detailed and comprehensive and not diverge/deviate from the corresponding lesson plans.

(CDRL A020, Student Outlines)

4.2.7 Media. The Contractor shall develop CBM+ operator and maintainer media in Microsoft Office PowerPoint format. The CBM+ operator and maintainer media shall correlate with the corresponding lesson plan and support the transfer of knowledge to the student.

(CDRL A021, Media)

4.2.8 Performance Evaluations. The Contractor shall develop CBM+ and FARS Dashboard performance evaluations. The performance evaluations shall be designed to evaluate the knowledge and skills required to master the objectives (TLO and ELOs). The Government shall review the performance evaluations and the Contractor shall correct and resubmit for final Government review and approval (CDRL A022, Performance Evaluations)

4.2.9 Training Verification. Training Verification will take place prior to I&KPT. The training verification will consist of reviewing the training deliverables to ensure the OEM is prepared for conducting I&KPT.

4.2.10 Instructor & Key Personnel Training. The Contractor shall provide Instructor & Key

Personnel Training. The I&KPT shall be developed and delivered to support the “train the trainer concept” to ensure that Marine Corps trainers obtain complete understanding of the operation, functionality, troubleshooting, and maintenance procedures. The Contractor shall deliver the operator course one time and the maintenance course one time to instructors and key personnel. Operator courses will be structured to have a maximum of 12 students and Maintenance courses will be structured to have a maximum of 12 students.

(CDRL A023, Training Delivery Requirements)

4.3 Provisioning

4.3.1 The Government will chair and the Contractor shall host the Provisioning Guidance Conference (PGC) within 60 days after contract award. The purpose of the meeting is to clarify requirements for Provisioning Technical Documentation (PTD) for the elements selected using GEIA-HB-0007, GEIA-STD-0007, the GEIA Attribute Selection Sheet, and MIL-HDBK-502 to include format and medium of delivery, addressing of unique Government requirements, and discussion/resolution of any unclear provisioning/cataloging related areas. Joint development of a PPS using Form MC-ALB- 4423/15 and scheduling of periodic PTD submissions for Government review shall be addressed and documented at this conference.

4.3.2 The Contractor shall host the Provisioning Conference(s) at the Contractor’s facility IAW the PPS. The Contractor shall provide copies of the Provisioning Parts List (PPL), Engineering Data For Provisioning (EDFP), and a production CBM+ hardware. The Contractor shall be responsible for disassembly of the equipment during this conference to validate and verify all PTD. CDRLs for associated deliverables are covered under the appropriate sections of this PWS.

4.3.3 The Contractor shall establish, manage, and submit a Provisioning Plan (PP). The objectives of the PP are to provide optimum material readiness, provide economical logistics support, and identify/evaluate resources required to develop and manage an effective support system. All design, modification/alteration, and engineering activity shall require PTD. MIL-HDBK-502 and GEIA-HB-0007 may be used for additional guidance. Provisioning will be accomplished down to the Field Maintenance Level to include calibration, removal and replacement of LRU, and repair of non-molded cables and/or connectors. Provisioning status, identification of problem area(s), and necessary resolutions to problems shall be addressed as soon as identified.

(CDRL A026, Provisioning Plan (PP))

4.3.4 The Contractor shall develop/document Provisioning Technical Documentation (PTD) to include, but not be limited to, a Provisioning Parts List (PPL), Provisioning Parts List Index (PPLI), Long Lead Time Items List (LLTIL), Tools and Test Equipment List (TTEL), Common and Bulk Items List (CBIL), and any Design Change Notices (DCNs). The Government will concur in the format, medium of delivery, and frequency for submission of the PTD at the PGC.

4.3.5 Provisioning Parts List (PPL). The Contractor shall include the following in the PPL: the end item, component or assembly, and all Field level maintenance support items that can be disassembled, reassembled, or replaced, and which, when combined, constitute the end item, component, or assembly and shall include items such as parts, materials, attaching hardware, connecting cabling, piping, and fittings required for the operation and maintenance of the end item, component, or assembly. The PPL shall be used to determine the range and depth of support items required to maintain the end item for an initial period of service. This includes all commercial items, unless excluded by the provisioning requirements. The final PPL shall be updated using the screening results provided by Defense Logistics Agency - Logistics Information Services (Log Info Svcs).

(CDRL A027, Logistics Product Data Summaries – Provisioning Parts List (PPL))

4.3.6 Provisioning Parts List Index (PPLI). The Contractor shall deliver a PPLI that cross references the PPL from part number to Provisioning List Item Sequence Number (PLISN). The PPLI shall be in alpha/numerical part number sequence. The cross-reference shall be on every PLISN that appears in the PPL. The cross- reference shall include all primary part numbers as well as all secondary part numbers.

(CDRL A028, Logistics Product Data Summaries – Provisioning Parts List Index (PPLI))

4.3.7 Long Lead Time Items List (LLTIL). The Contractor shall provide a LLTIL that shall contain those items which, because of their complexity of design, complicated manufacturing process, or limited production capacity, may cause production or procurement cycles that would preclude timely and adequate delivery, if not ordered in advance of normal provisioning.

(CDRL B002, Logistics Product Data Summaries – Long Lead Time Items List (LLTIL)

4.3.8 Tools and Test Equipment List (TTEL). The Contractor shall provide a TTEL that shall contain those support items required to inspect, test, calibrate, service, repair, or overhaul an end item to the Field level of maintenance.

(CDRL B003, Logistics Product Data Summaries – Tools and Test Equipment List (TTEL))

4.3.9 Common and Bulk Items List (CBIL). The Contractor shall, at a minimum, indicate the material type, grade, and class for items that are difficult or impractical to list on a top down/disassembly sequence PPL, but for which provisioning is required to support the operation of the end item/equipment. The Contractor shall submit sufficient information to enable the Government to relate the material/ specification number to the pertinent item. These items are subject to wear or failure, or otherwise required for maintenance, including planned maintenance of the end item/equipment.

(CDRL B004, Logistics Product Data Summaries – Common and Bulk Items List (CBIL))

4.3.10 Design Change Notice (DCN). The Contractor shall use a DCN to identify changes to Provisioning Technical Documentation that add to, delete, supersede, or modify items previously listed that are approved for incorporation into the end item.

(CDRL B005, Logistics Product Data Summaries – Design Change Notice (DCN))

4.3.11 LSA-036 Pre-Screening. The Contractor shall provide, upon conditional acceptance of the PPL and IAW the PPS, a copy of the PPL to Log Information Services for the purpose of pre-screening. Pre-screening data is used to identify existing NSNs, validate currency of an NSN, validate part numbers and CAGE codes on a PPL, aid in maximum use of known assets, and confirm accuracy of provisioning data before it is submitted for cataloging. When the PPL is part of an incremental provisioning effort, all PLISNs that were previously pre-screened shall be identified to avoid duplication.

(CDRL B006, Logistics Product Data – LSA-036 Pre-Screening)

4.3.12 Engineering Data for Provisioning (EDFP). The Contractor shall provide Engineering Data For Provisioning, also referred to as form, fit, and functional data (as referenced in the Defense Federal Acquisition Regulation Supplement [DFARS]), is technical data used to describe parts/equipment and consists of data such as specifications; standards; drawings; photographs; sketches and descriptions; necessary assembly and general arrangement drawings; schematic drawings; schematic diagrams; and wiring and cable diagrams necessary to indicate the physical characteristics, location, and/or function of the item.

(CDRL B007, Engineering Data for Provisioning (EDFP))

4.3.13 EDFP shall NOT be delivered when the item is:

a. Identified by a Government specification or standard that completely describes the item, including its materiel, dimensional, mechanical, and electrical characteristics.
b. Identified in the Federal Logistics Information System (FLIS) with a fully described NSN; EDFP may be required if there are no U.S. users on an NSN, an NSN is cancelled, or an NSN upgrade isrequired.
c. Listed as a reference item (subsequent appearance of an item) on a parts list.

4.3.14 Items Logistics Data Record (ILDR). The Contractor shall deliver an ILDR IAW FED-STD-5, Preparation of Proposed Item Logistics Data Records in the event EDFP is not available or is inadequate.

(CDRL B009, Technical Report – Study/Services)

4.3.15 Spares Acquisition Integrated with Production (SAIP). The Contractor shall provide a SAIP listing of all LRUs present on the PPL. The LRU list shall reflect those items that can be removed and replaced at the Field level of maintenance. The list shall include, but is not limited to: description, manufacturer, manufacturer part number, CAGE code, price per unit, quantity per system, NSN (if applicable), and Maintenance Replacement Rate 1 (MRR1) (breakage per thousand per year). The Contractor shall recommend range and depth of LRU spares sufficient to support two (2) years of system operations. The Contractor shall also provide a list of refreshed consumables and anticipated rate of consumption.

(CDRL B008, Logistics Product Data Summaries – Spares Acquisition Integrated with Production (SAIP))

4.3.16 Spare Parts. The contractor shall provide spare parts in the quantities to be determined by the USMC Maintenance Strategy determination. The maintenance strategy will be determined by Marine Corps Systems Command using input from Marines, a Level of Repair Analysis, and recommendations from the SAIP.

4.3.17 Diminishing Manufacturing Sources and Material Shortages (DMSMS). The Contractor shall provide DMSMS forecasting source data to enable the identification, forecasting, and management of piece part obsolescence impacts and mitigations.

(CDRL B010, Obsolescence Alert Notice)

4.4 Maintenance

4.4.1 Firmware Updates. The Contractor shall provide access to any future firmware updates to the data logger hardware along with documentation with operational impact.

4.4.2 Software Installation and Re-Imaging. The Contractor shall ensure future software and firmware updates to the data logger's baseline will continue to allow Marine maintainers to install software.

4.5 Identification and Marking Requirements

4.5.1 Data Plate. Each data logger shall come with a system identification data plate showing the specific UID markings as defined in MIL-STD-130. The exact information to be provided on the data plate will be provided by the Government during the Post Award Conference. At a minimum the identification plate shall include the following information: TAMCN, Nomenclature, NSN, and serial number. The serial number range and format to be used will also be provided by the Government. In addition, each data plate shall be marked with a Unique Item Identifiers (UIIs). The identification plate shall be permanently affixed, in a readily accessible and visible location. Data plates shall conform to A-A-50271 or contractor's proposed plates subject to the approval by the Government.

4.5.2 Item Unique Identification (IUID). The Contractor shall establish UIIs by assigning a machine-readable character string or number to each data logger system IAW MIL-STD-130N, Change 1. Each data logger shall be marked with a UII. The Contractor shall assess the need to provide an IUID label to any associated components of the data logger system IAW MIL-STD-130N, Change 1. All UIIs shall be registered in the IUID Registry, and the UII must remain unchanged throughout the item's life, to include modification or upgrade. The Contractor shall provide an IUID Marking and Verification Report to provide the data resulting from IUID marking activities, such as: physical asset marking, registration, verification, inventory audits, quality audits, and other asset lifecycle activities, as applicable. All end items, components, sub-components, assemblies, sub-assemblies, and parts meeting IUID requirements (see DFARS 252.211-7003) shall be marked with IUID-compliant 2D Data Matrix symbols in the EDFP IAW the latest version of MIL-STD-130.

(CDRL B011, IUID Marking and Verification Report)

5 TRAVEL

5.0 Travel

5.0.1 Travel to other government facilities, other contractor facilities, CONUS and OCONUS may be required at the discretion of the government. All travel requirements (including costs, plans, agenda, itinerary, or dates) shall be pre-approved by the COR (subject to local policy & procedures).

5.0.2 Costs for travel shall be billed in accordance with FAR 31.205-46 Travel Costs. The government anticipates that additional travel may be identified over the life of the PWS. All travel will be in accordance with Join Travel Regulations.

Acronyms

API	Application Programming Interface
ASME	American Society of Mechanical Engineers
BIT	Built-in Test
CAGE	Contractor and Government Entity
CBIL	Common Bulk Item List
CDRL	Contract Data Requirements List
CM	Configuration Management
CSP	Calibration Service Provider
DCN	Design Change Notice
DFARS	Defense Federal Acquisition Regulation Supplement
DMSMS	Diminishing Manufacturing Sources and Material Shortages
DoD	Department of Defense
EDFP	Engineering Data For Provisioning
ELO	Enabling Learning Objective
EPA	Environmental Protection Agency
ESD	Electrostatic Discharge
ESDS	Electrostatic Discharge-Sensitive
ETM	Electronic Technical Manual
FDTM	Final Draft Technical Manual
FLIS	Federal Logistics Information System
FRC	Final Reproducible Copy
HMMP	Hazardous Materials Management Program
I&KP	Instructor and Key Personnel
I&KPT	Instructor and Key Personnel Training
IAW	In accordance with
ILDR	Items Logistics Data Record
IMP	Integrated Master Plan
IMS	Integrated Master Schedule
IPMR	Integrated Program Management Report
IUID	Item Unique Identification_
LLTIL	Long Lead Time Items List
LRU	Line Replaceable Unit
MAC	Maintenance Allocation Chart
MRR1	Maintenance Replacement Rate 1
MTBF	Mean Time Between Failure
NSN	National Stock Number
OEM	Original Equipment Manufacturer

PAC	Post Award Conference
PDP	Product Drawing Package
PGC	Provisioning Guidance Conference
PP	Provisioning Plan
PPL	Provisioning Parts List
PPLI	Provisioning Parts List Index
PLISN	Provisioning List Item Sequence Number
PPS	Provisioning Performance Schedule
PTD	Provisioning Technical Documentation
QASP	Quality Assurance Surveillance Plan
SAIP	Spares Acquisition Integrated with Production
SAT	Systems Approach to Training
SDD	Software Design Description
SM&R	Source Maintenance and Recoverability
SOCTM	Sign-off Copy Technical Manual
PWS	Statement of Work
SVD	Software Version Description
TLO	Terminal Learning Objective
TM	Technical Manual
TMCR	Technical Manual Contract Requirement
TTEL	Tools and Test Equipment List
UII	Unique Item Identifier
VSC	Vulnerability Scan Compliance